Noizu Direct

INTERNAL DOCUMENTATION

Payment System Explained Simply

How money flows in, how creators get paid,
and what's left to do before we go live.

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Part 1 — The Big Picture

Noizu Direct is a marketplace where **creators** (artists, makers, digital sellers) sell products to **buyers**. Money flows through the platform. We take a cut, release earnings to creators after a safety window, and pay them out in their local currency.

🛒 Buyer Pays

Buyer checks out using their card or Google Pay. We collect the money via Airwallex (our payment processor).

💰 Creator Earns

Creator's share sits safely in the platform. After the escrow window clears, it's ready for payout every Friday.

```
// Simple money math:
Buyer pays → Gross Amount
− Processing fee (Airwallex charge to us)
− Platform fee (our commission)
= Creator's Net ← what the creator actually earns
```

Part 2 — Payment IN (Buyer Pays)

How a buyer completes a purchase

1. **Buyer adds to cart & checks out**
   They pick a product, click Buy, and reach the checkout page.

2. **Currency is detected automatically**
   The platform detects the buyer's region and suggests the matching currency (e.g. MYR for Malaysia, SGD for Singapore, USD as default).

3. **Live FX rate is fetched**
   We call Frankfurter API to get the real USD → local currency rate. The buyer sees the price converted live.

4. **Payment widget loads (Airwallex Drop-in)**
   The buyer sees a payment form from Airwallex — card details, Google Pay, and eventually local methods like FPX (Malaysia) or PayNow (Singapore).

5. **Buyer pays**
   Airwallex processes the payment securely. We never touch raw card numbers — Airwallex handles PCI compliance.

6. **Webhook fires & order is created**
   Airwallex sends us a webhook saying "payment confirmed". Our server creates the Order and Transaction records in the database.

7. **Funds go into Escrow**
   The transaction is marked
   ESCROW
   . Money is held for a safety window (configurable — e.g. 7 or 14 days) before being released to the creator.

8. **Escrow clears → COMPLETED**
   After the window passes, the transaction moves to
   COMPLETED
   . The creator's net amount is now available for payout.

> ⏳ **What is Escrow and why do we use it?**
>
> Escrow is a temporary hold on funds. If a buyer disputes a charge or requests a refund within the window, we can return the money without chasing the creator. Once the window passes safely, the money is released. Think of it like a cooling-off period before the creator can take the money out.

Supported Payment Methods

METHOD	STATUS	NOTES

💳 Credit / Debit Card	LIVE	Visa, Mastercard, etc.
🍎 Google Pay	LIVE	Works on supported browsers/devices
🏦 FPX (Malaysia)	PENDING	Requires Airwallex to enable on our account — contact support
🏦 PayNow (Singapore)	PENDING	Same — requires Airwallex account enablement
🍎 Apple Pay	TBA	Possible via Airwallex Drop-in, needs domain verification

Why don't FPX/PayNow show yet? These local payment methods require Airwallex to activate them on your merchant account. This is an account configuration step — the code is already ready to handle them automatically once enabled.

Part 3 — Payment OUT (Creator Gets Paid)

How creators receive their earnings

1. **Creator checks their Earnings dashboard**
   Shows Available Balance, In Escrow, Total Earned, and Total Paid Out. All updated in real-time.

2. **Creator sets their payout currency (one-time setup)**
   In their Profile settings, the creator selects what currency they want to receive (e.g. MYR, SGD, IDR, USD). This is locked until they change it again.

3. **Every Friday at 10am Malaysia Time — automated sweep runs**
   A scheduled job (GitHub Actions → our server) automatically checks all creators with cleared earnings.

4. **Minimum threshold check: USD 10**
   If a creator's available balance is less than USD 10 (or equivalent), the payout is skipped that week and accumulates for next Friday.

5. **Payout is calculated & sent via Airwallex**
   We call Airwallex's transfer API. The creator's USD earnings are converted to their chosen currency at the live rate and sent to their bank.

6. **Creator receives a payout email**
   Notification sent with the amount and status. Payout appears in their dashboard history.

7. **Reconciler runs & updates status**
   A second job checks Airwallex for the transfer outcome and marks payouts as PAID, PROCESSING, or FAILED accordingly.

```
// Payout calculation:
Available = Total COMPLETED earnings - Total already paid out

// Minimum check:
If Available < USD 10 → skip, roll over to next Friday

// Currency conversion:
Payout Amount = Available (USD) × Live FX rate → Creator's currency
```

Payout Schedule

WHEN	WHAT HAPPENS
Every Friday 2:00 AM UTC *(10:00 AM Malaysia time)*	Automated payout sweep runs for all eligible creators
Within minutes	Airwallex transfer initiated, creator emailed

Shortly after	Reconciler checks transfer status, updates records
1–3 business days	Money lands in creator's bank (varies by bank/country)

✅ What counts as Available

Transactions in **COMPLETED** status only. Escrow transactions are not included until they clear.

⏳ What's still in Escrow

Transactions in **ESCROW** status. Visible on the earnings dashboard but not payable yet.

Monthly Statement

Creators can view a full monthly breakdown at **Dashboard** → **Statement**:

- Every sale, fee, and net amount — grouped by month
- Every payout — grouped by the month it was requested
- Toggle between USD and their local payout currency
- Download as CSV for their accountant

Part 4 — Fees & Policies

Fee Structure

FEE TYPE	WHO PAYS	NOTES
Processing Fee	Deducted from gross	Airwallex's charge for handling the transaction (varies by currency/method)
Platform Fee	Deducted from gross	Noizu Direct's commission — configured per transaction type
Creator's Net	Creator receives	Gross − Processing Fee − Platform Fee
FX Spread	Implicit in rate	Live rate used; actual payout rate locked at Airwallex transfer time

Transaction Statuses

STATUS	MEANING	PAYABLE?
ESCROW	Payment received, holding for safety window	No
COMPLETED	Escrow cleared, funds available	Yes
PENDING	Payment being processed	No
FAILED	Payment failed or reversed	No
REFUNDED	Buyer refunded	No

Payout Statuses

STATUS	MEANING
PENDING	Payout created, queued for transfer
PROCESSING	Airwallex transfer initiated, in transit
PAID	Money delivered to creator's bank
FAILED	Transfer failed — requires investigation

Part 5 — What's Done ✅ & What's Pending ⬜

✅ Completed & Working

✅ **Airwallex Drop-in checkout widget**
Card payment + Google Pay live. Currency-aware (buyer sees price in their local currency).

✅ **Multi-currency checkout**
Auto-detects region, fetches live FX rates, lets buyer change currency without losing their cart or address details.

✅ **Webhook → Order + Transaction creation**
When Airwallex confirms payment, our server records the order, sets status to ESCROW, splits fees correctly.

✅ **Escrow tracking**
Funds held until configurable window clears, then status moves to COMPLETED automatically.

✅ **Creator Earnings dashboard**
Shows Available Balance, In Escrow, Total Earned, Total Paid Out. Real-time from database.

✅ **Monthly Statement page with CSV export**
Full breakdown by month. Buyer names, product titles, fees, net. USD ↔ local currency toggle. Downloadable CSV.

✅ **Automated weekly payout cron (code complete)**
Friday 10am MYT sweep. USD 10 minimum threshold. Groups by creator, converts to payout currency, transfers via Airwallex.

✅ **GitHub Actions scheduler**
Cron job configured. Triggers payout sweep + reconciler every Friday automatically. Manual trigger also available.

✅ **Payout reconciler**
Checks Airwallex transfer status after payout, updates PENDING → PAID / FAILED in database.

✅ **Creator payout currency selection**
Creator sets preferred payout currency once in their profile. Used for all future payouts.

✅ **Zero-decimal currency support**
IDR, JPY, KRW, VND handled correctly (no cents — just whole units).

✅ **Payout email notification**
Creator receives email when payout is initiated.

⬜ Pending / TBA Before Go-Live

🚀 **Deploy code to Vercel production**

Both commits (multi-currency checkout + payout system) are on master branch but not yet deployed. GitHub Actions 404s until this is done.

BLOCKER

🔗 **Update GitHub Secret: APP_URL → main domain**

Currently set to `https://noizu-direct.vercel.app`. Once you switch to `noizu.direct` as primary domain, update this secret in GitHub → Settings → Secrets → Actions → APP_URL.

TBA — domain migration

🏦 **Enable FPX (Malaysia) on Airwallex account**

Contact Airwallex support and request FPX activation on your sandbox merchant account. Code is ready — it appears automatically once enabled.

TBA — Airwallex support

🏦 **Enable PayNow (Singapore) on Airwallex account**

Same process as FPX. Request activation from Airwallex. Code handles it automatically.

TBA — Airwallex support

🧪 **End-to-end payment test (sandbox)**

Use Airwallex sandbox test card → complete checkout → verify webhook fires → check order in database → confirm transaction created with correct status → wait for escrow → verify COMPLETED.

TEST BEFORE LIVE

🧪 **Test payout cron manually after deploy**

After deploy, run the GitHub Actions workflow manually (Actions → Weekly Payout Sweep → Run workflow). Check logs confirm 200 OK. Verify payout records created in database.

TEST AFTER DEPLOY

🧪 **Test USD 10 minimum threshold**

Create a transaction with less than USD 10 net. Run payout sweep. Confirm creator is skipped. Create one with more. Confirm payout is triggered.

TEST AFTER DEPLOY

🔐 **Verify Airwallex webhook secret is set in production**

The `AIRWALLEX_WEBHOOK_SECRET` environment variable must be set in Vercel production environment for webhooks to be verified correctly.

CHECK ON DEPLOY

📊 **Admin view of all payouts & creator balances**

Currently creators see their own data. An admin panel to see all creators, their balances, and trigger manual payouts is not yet built.

FUTURE

🛡️ **Dispute & refund flow integration with payouts**

If a buyer raises a dispute, the escrow window helps, but a full refund-reversal flow that claws back from creator payouts is not yet implemented.

FUTURE

🌐 **Migrate custom domain noizu.direct as primary on Vercel**

Set `noizu.direct` as the production domain in Vercel. Update GitHub Secret APP_URL. Update Airwallex webhook URL in Airwallex dashboard to point to `https://noizu.direct/api/webhooks/airwallex`.

TBA — domain migration

Part 6 — Deploy Checklist

When you are ready to go live, do these steps in order:

1. **Deploy to Vercel**
   Push triggers auto-deploy, OR go to Vercel dashboard → noizu-direct → Deployments → Promote to Production.

2. **Verify all Vercel environment variables are set**
   Check: `AIRWALLEX_API_KEY`, `AIRWALLEX_CLIENT_ID`, `AIRWALLEX_WEBHOOK_SECRET`, `CRON_SECRET`, `DATABASE_URL`, `NEXTAUTH_SECRET`, `NEXT_PUBLIC_AIRWALLEX_ENV=prod`

3. **Update Airwallex webhook URL**
   In Airwallex dashboard → Webhooks → update endpoint to `https://noizu.direct/api/webhooks/airwallex`

4. **Update GitHub Secret APP_URL**
   GitHub → repo Settings → Secrets & Variables → Actions → APP_URL → update to `https://noizu.direct`

5. **Run the payout workflow manually to test**
   GitHub → Actions → Weekly Payout Sweep → Run workflow. Check logs for 200 OK.

6. **Make a test purchase with a real card (small amount)**
   Verify: payment succeeds → order created → transaction in ESCROW → email received.

7. **Contact Airwallex to enable FPX and PayNow**
   Provide your merchant account ID. They activate within a few business days. No code change needed.

⚠️ **Before going live with real payments:** Make sure `NEXT_PUBLIC_AIRWALLEX_ENV` is set to `prod` in Vercel (not `demo` or `sandbox`). Sandbox payments are test-only and will not process real money.